

Predictive Maintenance of Turbofan Engines using Random Forest Regression for Remaining Useful Life Estimation

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Abstract—Effective predictive maintenance is critical to the aviation industry to ensure safety and reduce operational costs. This paper presents a data driven approach to estimating the Remaining Use Life of turbofan engines. The study implements random forest regression to handle complex fault signatures. This research analyses all 4 sub datasets (FD001 – FD004) with ranging conditions from sea level to complex flight envelopes with multiple fault modes. Our experimental results demonstrates accurately that Random Forest model significantly outperforms the linear baseline, we get a Root Mean Square Error of 12.02 in baseline dataset FD001 and 20.25 on the highly volatile dataset that is FD004. Feature importance shows us critical shift in sensor dominance, which also confirms that the model has successfully filtered environmental noise to identify true mechanical degradation, also while acting in accordance with the thermodynamic principles.

I. INTRODUCTION (HEADING 1)

This paper presents a data-driven approach to estimate the Remaining Useful Life (RUL) of turbofan engines using the NASA C-MAPSS dataset. The data was used from an engine model which is a 90,000 lb. thrust class turbofan with five rotating components [1]. In the world of aviation, the concept of predictive maintenance is crucial, this can be an effective method to reduce costs and also avoid catastrophic failures. We cannot wait for a machine to actually break, and hence we update the data to a virtual model such as the one we designed and simulating run-to-failure degradation trajectories and come up with a solution as to when the wear is at its max and we may need to replace the part. The Study compares a baseline Linear Regression model against a Random Forest Regressor. Data processing techniques such as sensor selection based on variance and piecewise linear RUL labelling were implemented to enhance model performance. The results of this paper will demonstrate that the Random Forest model achieved a Root Mean Squared Error (RMSE) of approximately 17.03 cycles on the FD001 dataset and 17.06 cycles on the complex FD002 dataset significantly outperforming the linear baseline [2].

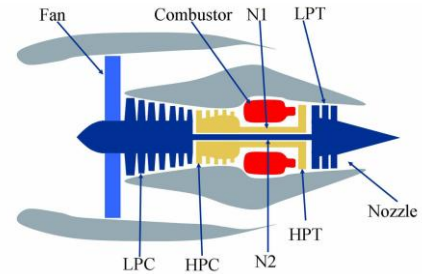


Figure 1. Simplified diagram of the 90K engine simulated in CMAPSS [1]

II. DATA PROCESSING

A. Dataset Description (Heading 2)

The study was conducted on dataset provided by NASA called the Commercial Modular Aero-Propulsion System Simulation. The data is divided into 4 categories, they range from FD001 through FD004. The data is from a fleet of engines of the same type, and every engine comes with a different degree of wear, manufacturing variation but these are not considered as fault condition. Furthermore, the data contains 3 different types of settings which can affect engine performance, and the data is contaminated with sensor noise.

B. Feature selection and Cleaning Data

The raw sensor data file contained non informative features that could introduce noise and increase computational complexity. Steps were taken to ensure that we filter such features:

- **Removal of Constant Sensors or Zero Variance Features:** A variance analysis was done on the dataset to identify which sensors had zero variance across the lifespan of the engine and using this custom variance we determined that sensors such as S_1, S_5, S_10 etc. were “dead sensors”. These sensors were removed as they provided no informational value to the model.
- **Correlation Analysis:** Following the removal of non-informative sensors, we moved on to implement Pearson correlation analysis on the

remaining sensors and confirm which sensor exhibited strong positive or negative correlation with the Remaining Use Life (RUL) and those with high positive correlation were retained as primary predictors.

- Clipping of data: In the early stages of the engine's life cycle any degradation is negligible, predicting a linear decline from the first cycle is physically inaccurate and can degrade the model performance. To address this inconsistency, we have introduced a threshold of 125 cycles. Any RUL value exceeding 125 was clipped to this limit. With this the model can consider the engine with more than 125 cycles as a fully healthy one and focus on the active degradation phase.

III. METHODOLOGY

Here we discuss the methods and models employed to predict the engine breakdown across the entire dataset, the output that was produced.

A. Baseline approach: Linear Regression

Initially the approach employed a standard Linear Regression (LR) model. This model assumed a direct linear relationship between the sensor readings and the Remaining Useful Life (RUL) across the entire dataset. The RMSE produced with this approach was considered as the baseline for future model, the model worked with data which did not go through any domain specific preprocessing and this is when we got a Baseline RMSE of 50.63 cycles.

B. Piecewise Linear RUL (Clipping)

The baseline model indicated high error rates, this was due to the model attempting to predict degradation in healthy engines. To correct this, the Piecewise Linear RUL strategy described in section II-C was applied. The model was retrained on this clipped data set. The new improved Linear Regression model produced results with an RMSE of 21.33 cycles

C. Non-Linear modelling: Random Forest Regressor

The Linear regression model captures general trends; however, engine degradation is a complex, nonlinear process influenced by interactions between varying pressure and temperature systems. To capture this complex curve shaped trend, we implemented Random Forest Regressor. The random forest is an ensemble method to Deep Neural network models, this model generates predictions from multiple decision trees to reduce variance and overfitting.

The specific model that was used in our prediction was configured with $n_estimators = 100$ and $max_depth = 15$, and was trained on the clipped data for predictive accuracy.

IV. EXPERIMENTAL RESULTS

A. Performance Metric

The models were evaluated using the Root Mean Squared Error (RMSE). This method penalizes larger prediction errors and provides a metric in the same unit as the target variable (Cycles).

B. Sensors used within the model to predict

Below is a table which contains every sensor data that was used to predict the cycle and how the engine degrades over time.

Symbol	Description	Units
s_1	Total temperature at fan inlet	°R
s_2	Total temperature at LPC outlet	°R
s_3	Total temperature at HPC outlet	°R
s_4	Total temperature at LPT outlet	°R
s_5	Pressure at fan inlet	psia
s_6	Total pressure in bypass-duct	psia
s_7	Total pressure at HPC outlet	psia
s_8	Physical fan speed	rpm
s_9	Physical core speed	rpm
s_10	Engine pressure ratio (P50/P2)	--
s_11	Static pressure at HPC outlet	psia
s_12	Ratio of fuel flow to Ps30	pps/psi
s_13	Corrected fan speed	rpm
s_14	Corrected core speed	rpm
s_15	Bypass Ratio	--
s_16	Burner fuel-air ratio	--
s_17	Bleed Enthalpy	--
s_18	Demanded fan speed	rpm
s_19	Demanded corrected fan speed	rpm
s_20	HPT coolant bleed	lbm/s
s_21	LPT coolant bleed	lbm/s

As we explored in section III Methodology, we saw how we progressed from baseline model to a decision tree based, nonlinear mode that is the Random Forest Regressor. We will now look into the results from 4 datasets below that FD001 – FD004.

a) *Dataset FD001*: The dataset FD001 was recorded in condition one (Sea Level) and has one fault modes that is the HPC Degradation. When we passed the dataset through our model, we found the following results.

The RMSE was 17.02 Cycles and when we retrieved the features that had the most importance with the final output we got the following output.

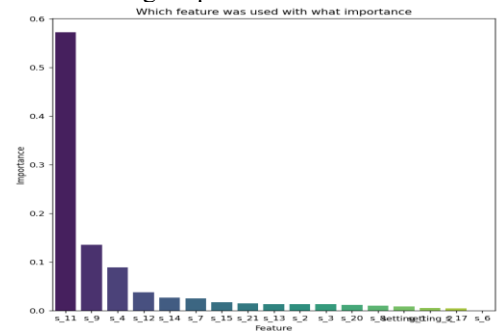


Fig 2. A feature importance map of the sensors which were primarily favored for the output

The map suggested that S_11 was the most important sensor for prediction along with S_9, S_4. Here S_11 is the static pressure at HPC outlet, while the others S_9 being physical core speed and S_4 being Total temperature at LPT outlet.

1. *Primary Indicator (S_11- static pressure at HPC Outlet)*: Since the fault is localized to the High-Pressure Compressor, the static pressure at the compressor's discharge is the most direct physical metric of its health. As the HPC degrades due to instances such as erosion in blade or due to increased tip clearance, its compression efficiency drops, causing observable deviations in S_11. Hence, we can scientifically be sure that the model identified the most critical feature.[2]
2. *S_9 – Physical core speed*: As the compressor degrades the thrust needed also increases, and the engine control system will compensate this by increasing the core rotational speed. Hence high importance of this sensor suggests the model learned to detect this compensatory stress.[3]
3. *Thermodynamic Stress (S_4 – Total temperature at LPT Outlet)*: As the efficiency of the engine drops, the exhaust gas temperature (EGT) rises. This is downstream proxy used by maintenance crews often called as “EGT Margin” [2].

b) *Dataset FD002*: This dataset unlike FD001 was not recorded at sea level but in condition six. The sensors that were highly favored while the model was applied on this new dataset were S_15, S_13, S_11, S_4. Here we see the core being clogged and not being able to pull in air efficiently and hence we see the Bypass Ratio (S_15) rise up significantly, as the air that passes through the engine chooses the path with the least resistance and this is through the Bypass duct. As the core becomes weak the computer forces the fan to spin faster and this where we see the spike in sensor S_13 which is associated with Corrected Fan Speed. We see a spike within S_11 and S_4 and these are due to the reasons mentioned in our previous result for the dataset FD001. The model predicts the data for FD002 with an RMSE of 17.06 Cycles.

c) *Dataset FD003*: With this dataset we get the RMSE of 20.7 Cycles. Here the most important sensor used to predict was S_11. In this dataset, we see that there are two fault modes that are HPC Degradation and Fan Degradation. We have seen S_11 sensor being prioritized in past datasets, however, here we see the S_11 being 4 times more important than other sensors. When the Fan and HPC both are low, that is when S_11 is the sensor with a high feature importance score. Furthermore, we see S_9 sensor being given importance, which is for engine core speed. As the thrust becomes low due to HPC failure and fan speed failure we see the core speed increasing to compensate this.

d) *Dataset FD004*: This is the most challenging of all datasets as it has six operating conditions and two fault modes, however, the model achieved RMSE of 20.25. This indicates strong generalization capabilities despite the high variance in the input. We also observe a shift in feature importance compared to the baseline datasets such as FD001 or FD003. While Static Pressure at the HPC Outlet S_11 was a dominant indicator in sea level conditions, corrected fan speed S_13 emerged as the overwhelmingly significant feature in FD004 with an importance score of approximately 3 times higher than the next closest feature.

In FD004 the aircraft changes altitude and the ambient pressure changes caused by climbing or descending create ‘noise’ that masks the subtle pressure drops caused by the fault. Hence the S_11 sensor was not significantly considered for this as it was in sea level tests.

In FD004 the model correctly identifies S_13 as the most significant sensor for the prediction because it is a corrected parameter. In aviation corrected speed is normalized to remove the effects of inlet temperature and pressure variations.

CONCLUSION

To conclude, this paper was aimed at predicting RUL in turbofan engines by taking a data driven approach. We also found that a linear approach is not an ideal approach when dealing with engine degradation and this when we moved to a non-linear approach with the Random Forest model. As we saw with Linear regression model where in the RMSE was approximately 50 cycles, however, with the Random Forest model we got the RMSE down to approximately 17 cycles. We also found the model did not just guess data but learned the physics behind the numbers. The models feature importance ranking aligned with the thermodynamic principles. We saw this with the correctly identified pressure sensor S_11 in dataset FD001 and FD003 as well as the correctly identified fan speed issue within the datasets FD002 and FD004. This model proves that maintenance of such critical machinery can be done with the help of our “condition-based maintenance” model, instead of using the schedule-based maintenance model, which would save millions in costs as well as accurately predict parts before they go into critical failure state. This model also stands as a proof that physics informed machine learning models are robust enough for real world deployment.

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